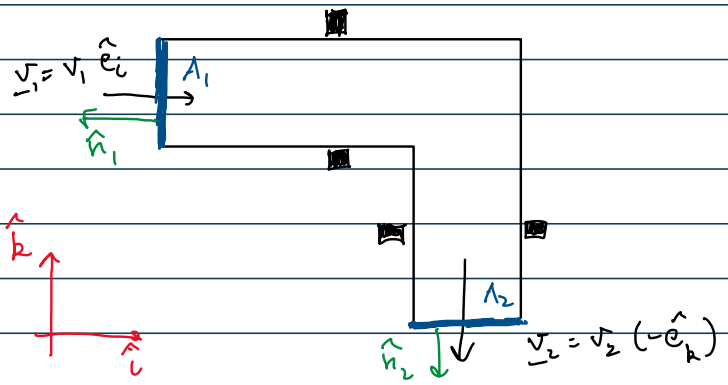
L6 Surface Tension, Reynolds Transport Theorem

Monday, 4 September 2023 10:11 PM

L6 Surface Tension, ...



+ Constant velocity
+ Normal vectors on each surface/opening

$\hat{n}_1 = -\hat{e}_i$

$\hat{n}_2 = -\hat{e}_k$

→ Momentum RTT:

w/ no s.s. $\quad \Sigma F = \iint_A \rho \underline{v} (\underline{v} \cdot \hat{n}) \, dA$

$$\Sigma F = \rho \underline{v}_1 \iint_{A_1} [(v_1 \hat{e}_i) \cdot (-\hat{e}_i)] \, dA + \rho \underline{v}_2 \iint_{A_2} [v_2 (-\hat{e}_k) \cdot (-\hat{e}_k)] \, dA$$

$\underline{v}_1 = v_1 \hat{e}_i$, $\hat{n}_1 = -\hat{e}_i$, $\hat{e}_i \cdot \hat{e}_i = 1$; $\underline{v}_2 = v_2 (-\hat{e}_k)$, $\hat{n}_2 = -\hat{e}_k$, $\hat{e}_k \cdot \hat{e}_k = 1$

$$= \rho (v_1 \hat{e}_i)(-v_1) \iint_{A_1} dA + \rho (v_2 (-\hat{e}_k)) v_2 \iint_{A_2} dA$$

$$\boxed{\Sigma F \Rightarrow -\rho v_1^2 A_1 \hat{e}_i - \rho v_2^2 A_2 \hat{e}_k}$$

7.